

Advanced Computer Graphics

Computer Animation

Interpolation-based Animation II

Interpolating Objects

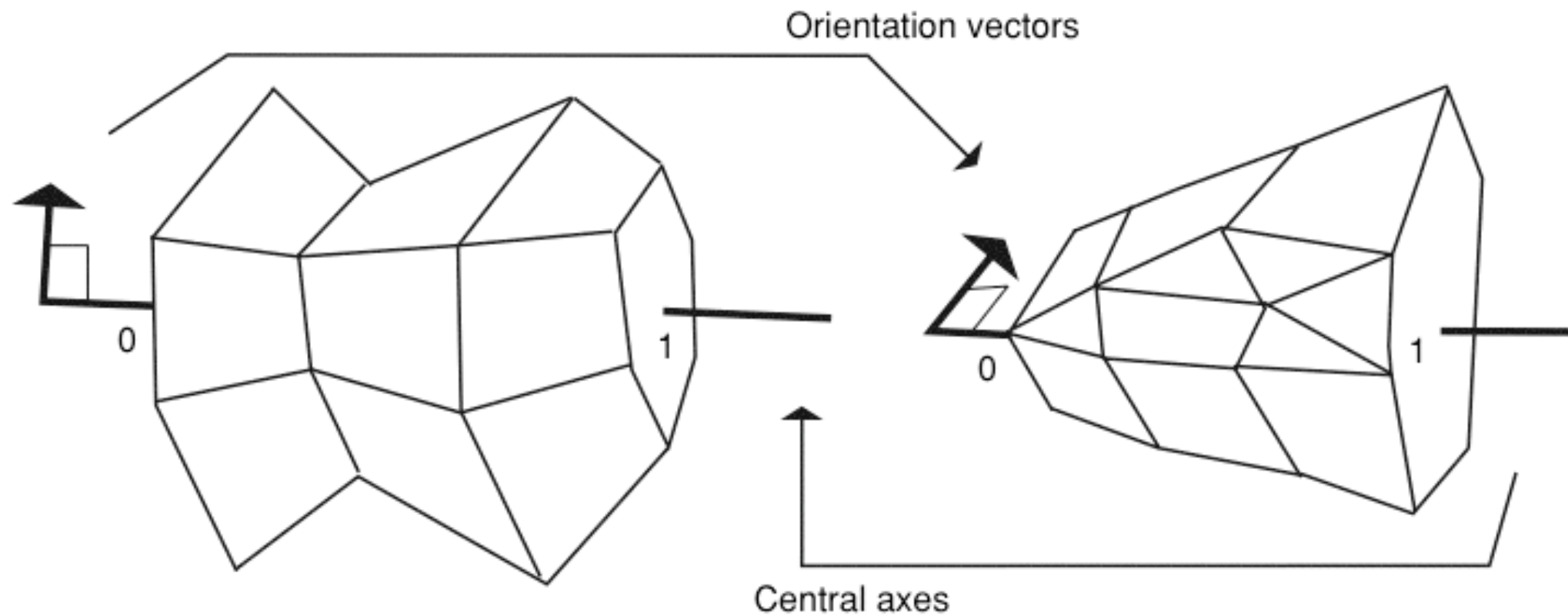
Interpolate Between 2 Objects

- Correspondence Problem
 - How to find?
- Objects with different topology / shapes
 - How to map?
- Surface-base approaches
 - Slice along one dimension
 - interpolate in other two maps both to sphere
 - Recursively divide into panels
- Volumetric approaches: Remeshing

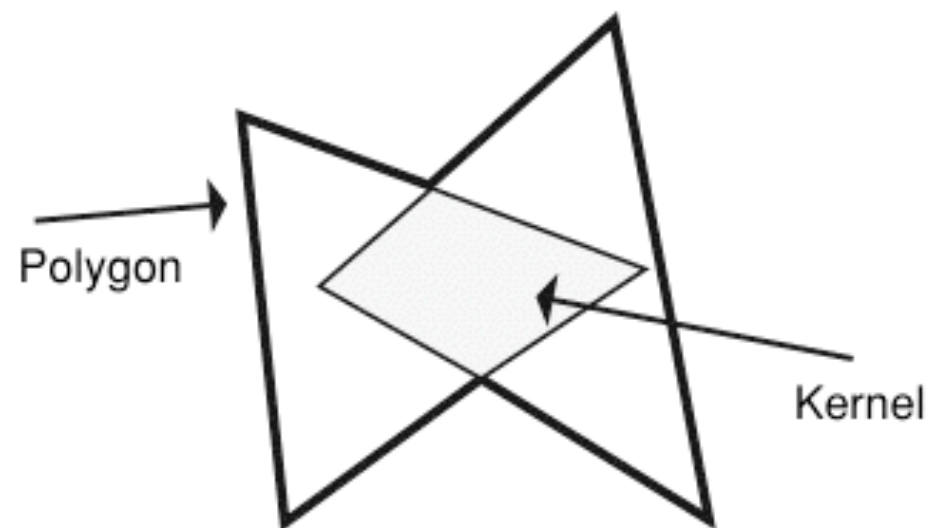
Case of Matching Topology

Object interpolation

For cylinder-like objects

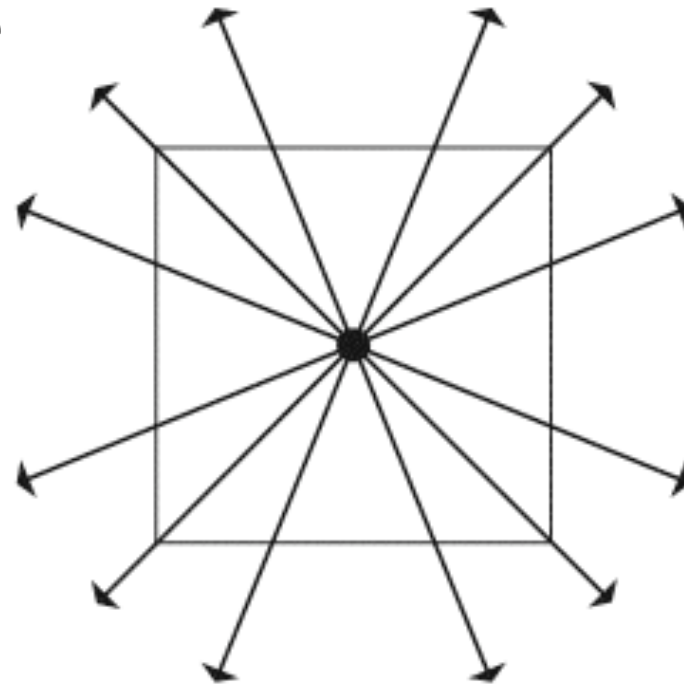


Radial mapping

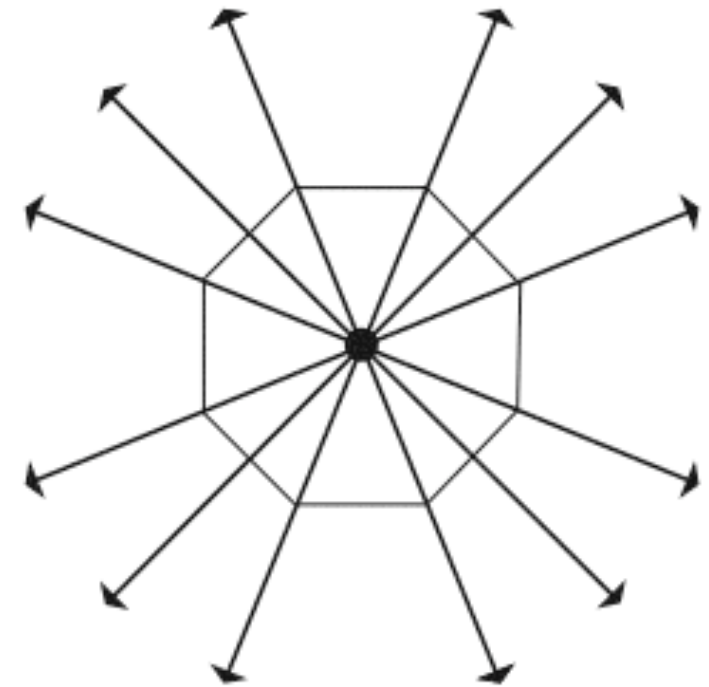


**If central axis intersects polygonal slice inside kernel
Then simple radial mapping possible**

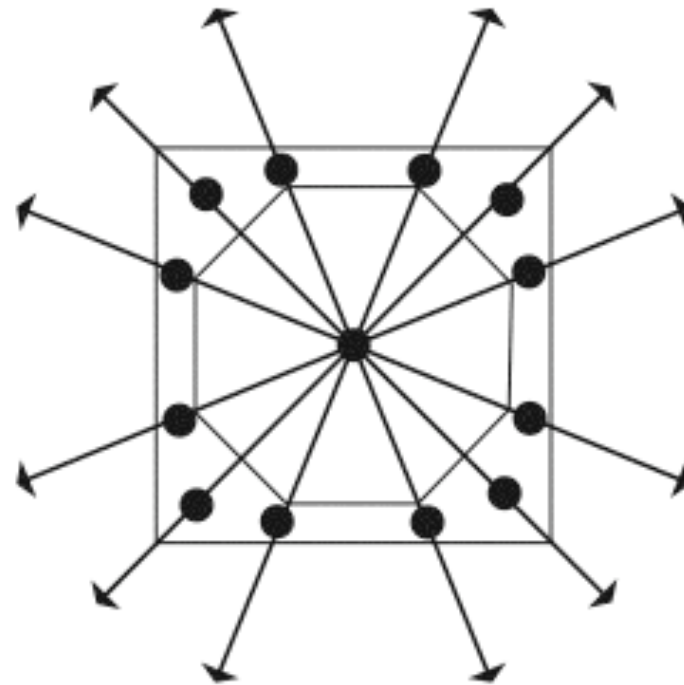
Object interpolation



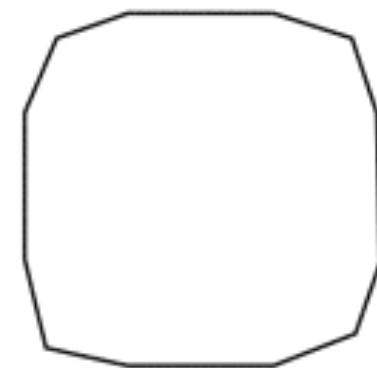
Sampling Object 1 along rays



Sampling Object 2 along rays

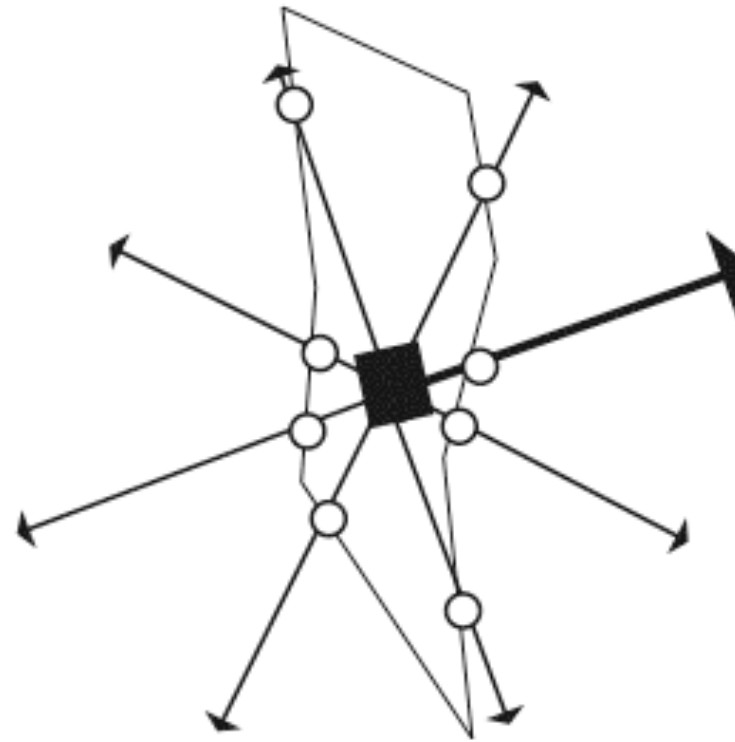
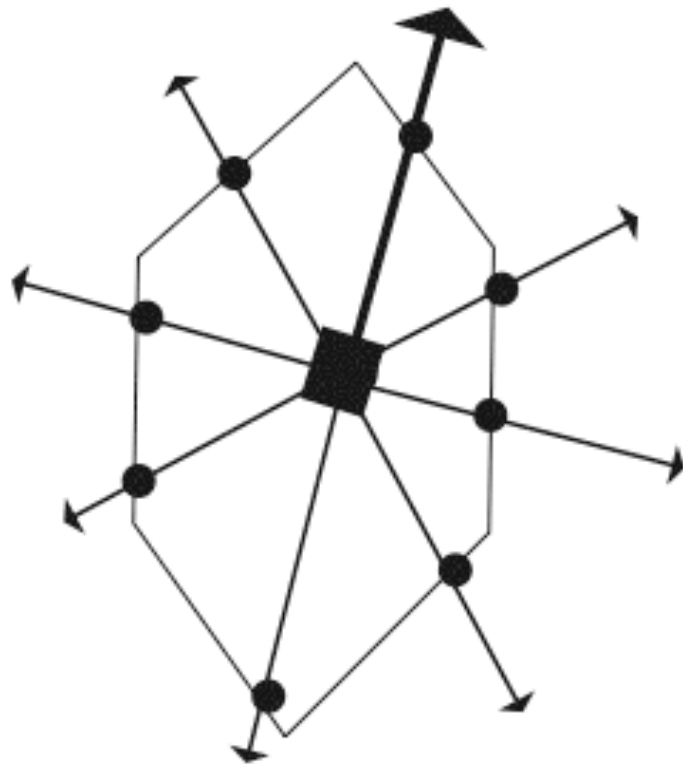


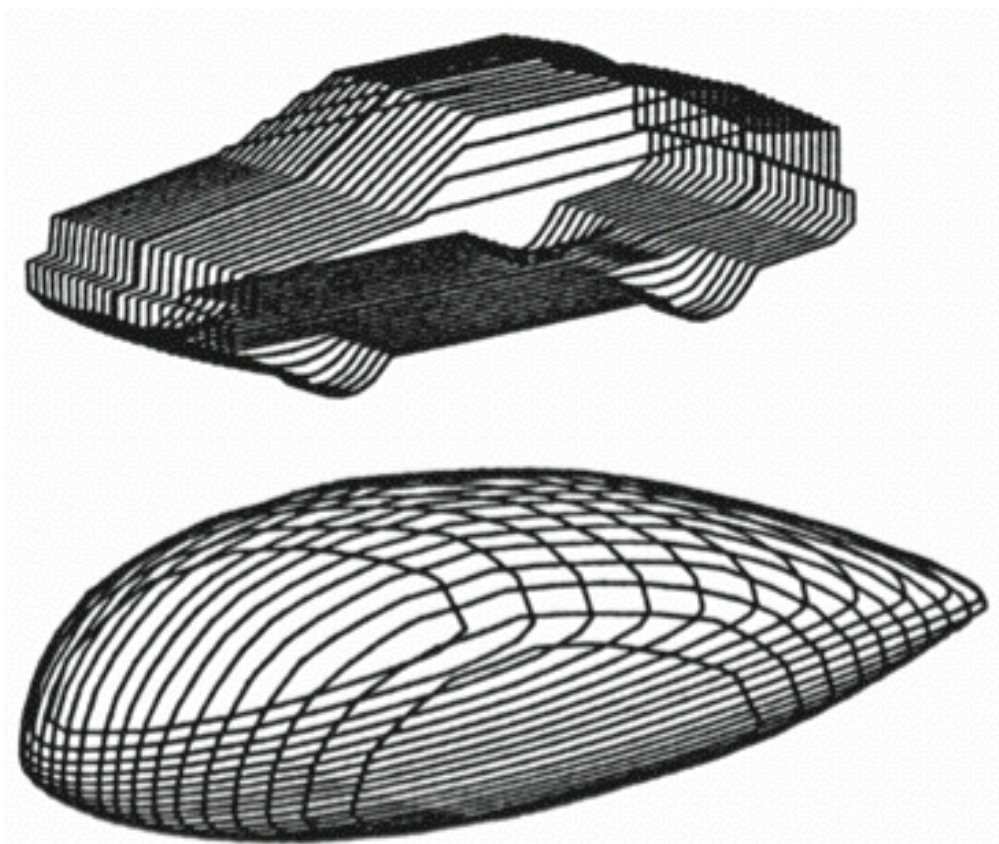
Points interpolated halfway between objects



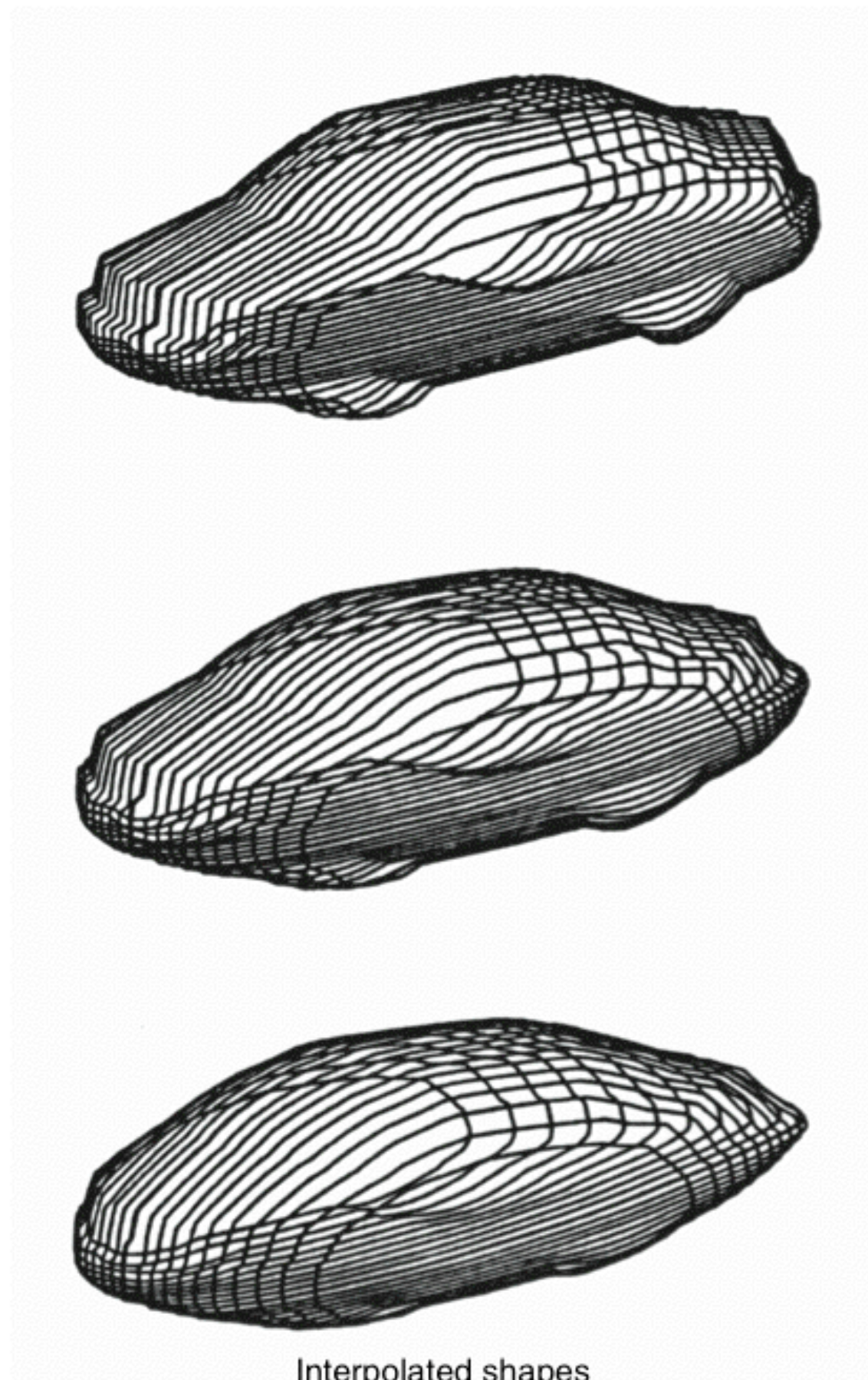
Resulting object

Object interpolation





Original shapes sliced into contours



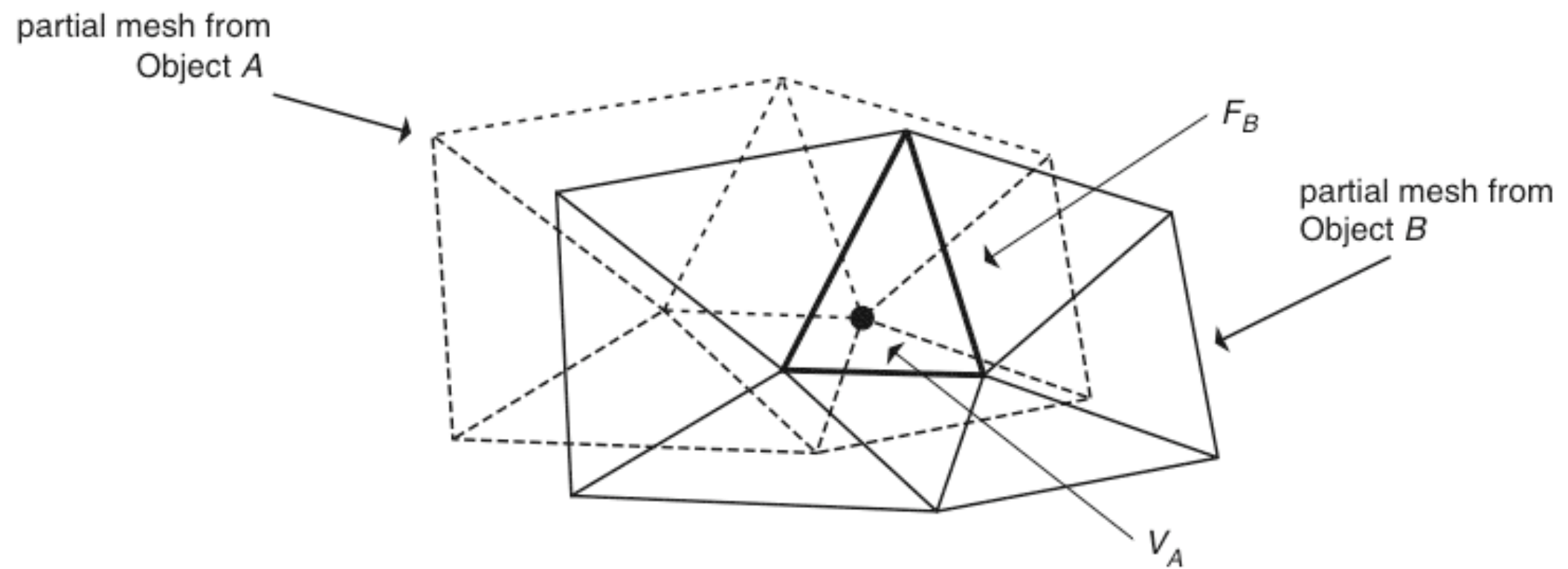
Interpolated shapes

Object interpolation for more complicated shapes

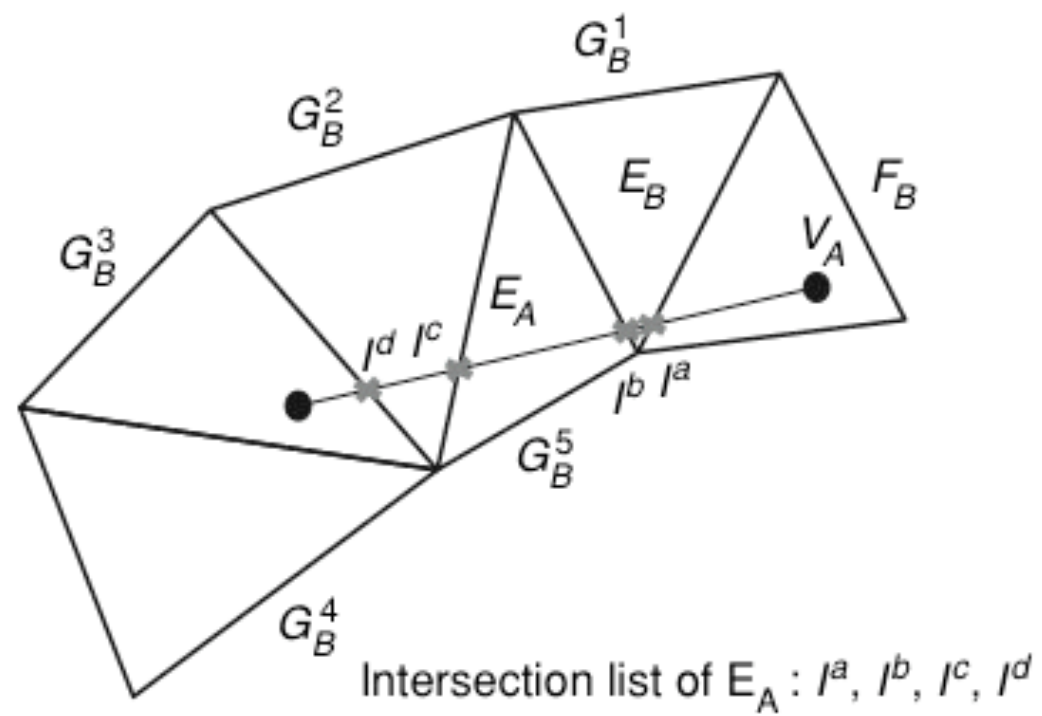
Spherical mapping to establish matching edge-vertex topology

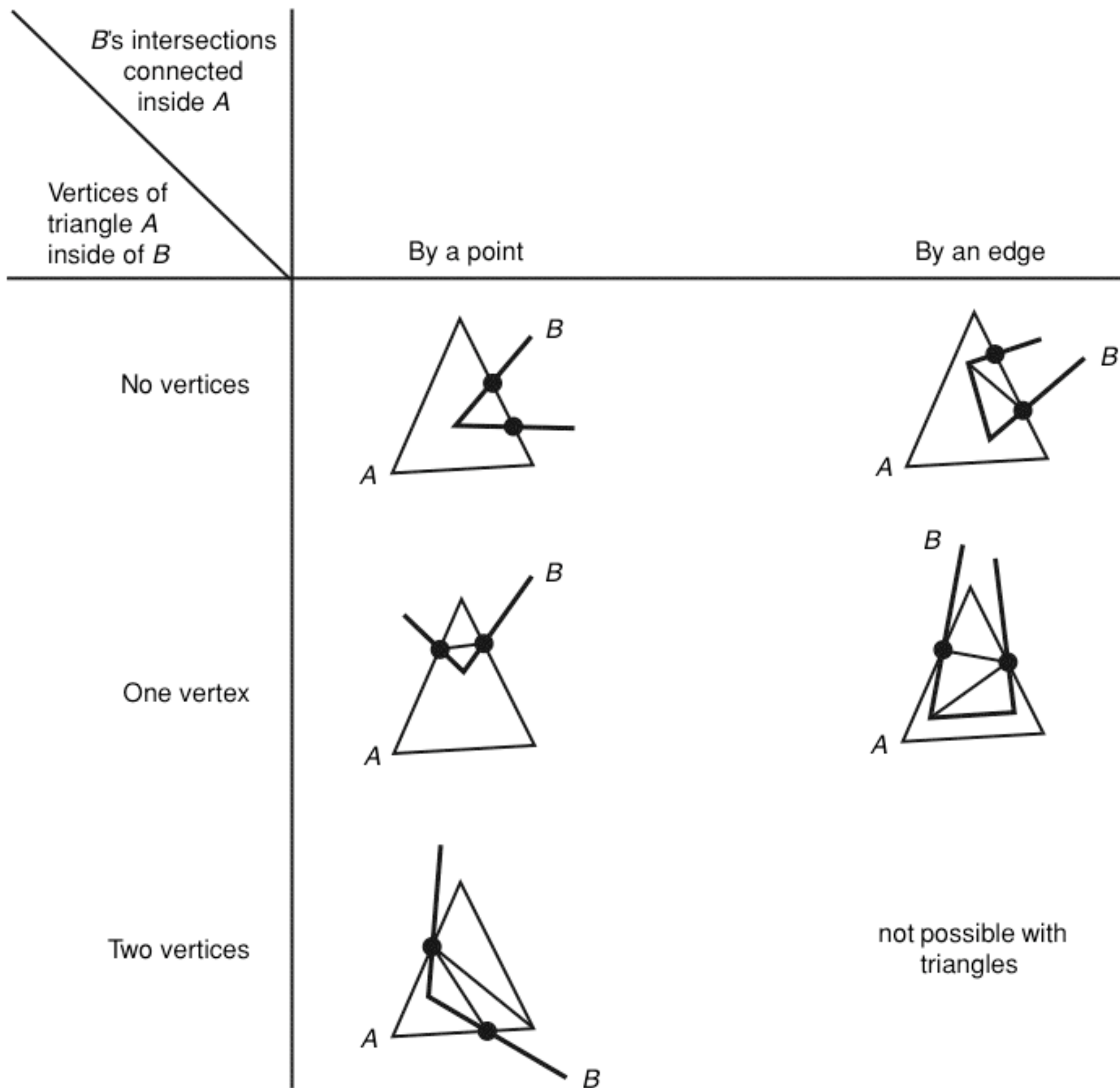
1. Map to sphere
2. Intersect arc-edges
3. Retriangulate
4. Remap to object shapes
5. Vertex-to-vertex interpolation

Map to sphere

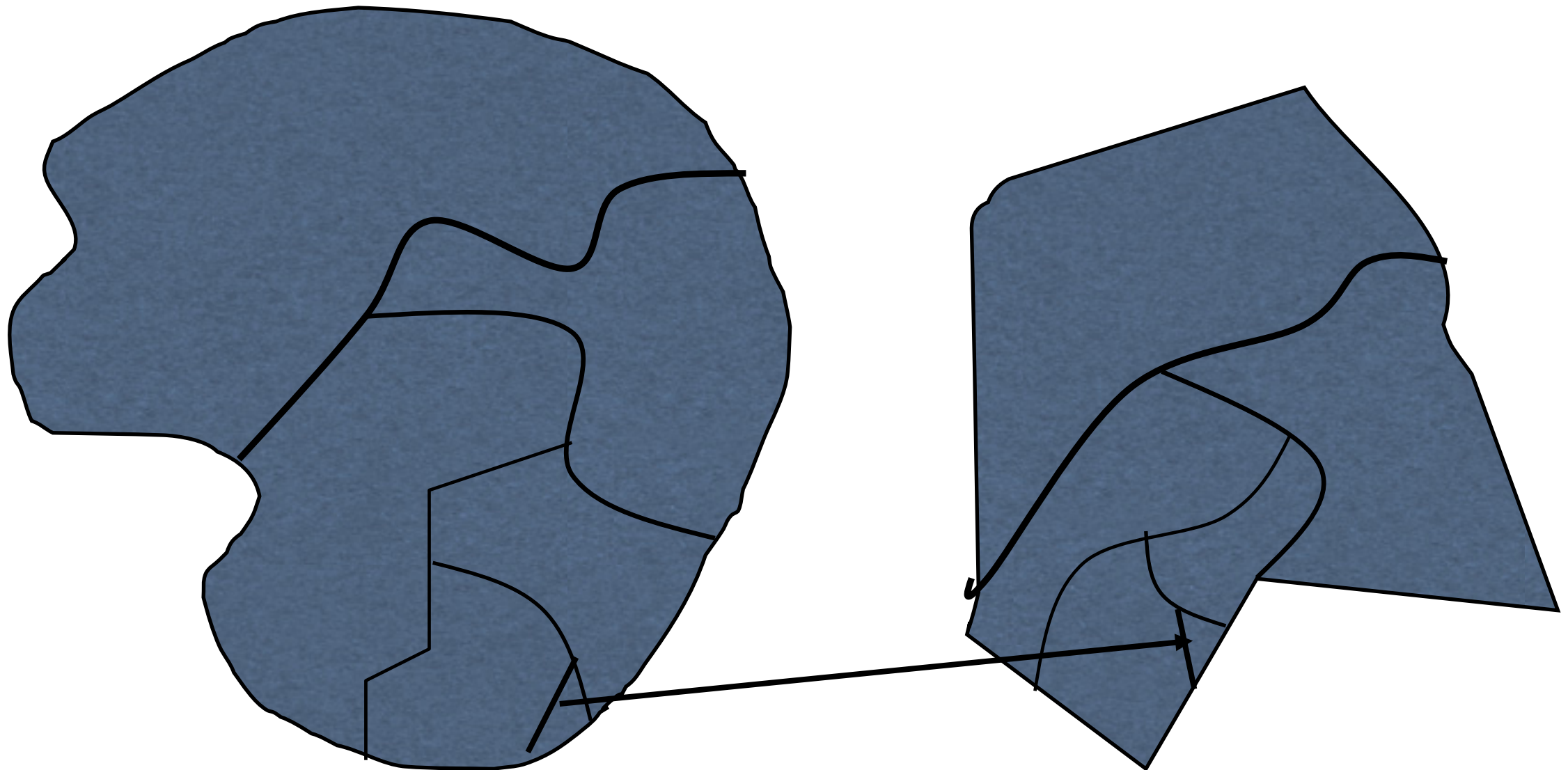


Object interpolation





Object interpolation - recursive sheets



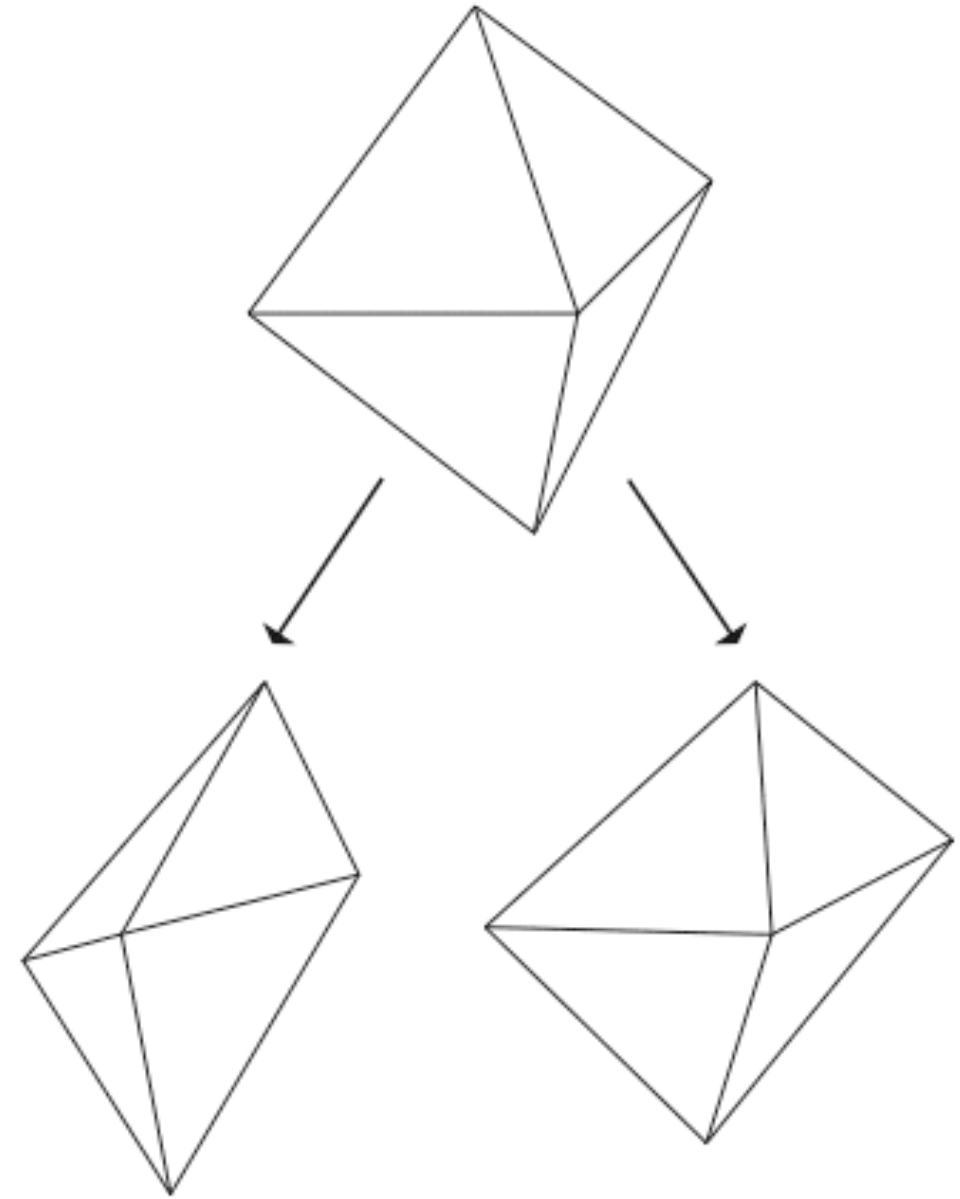
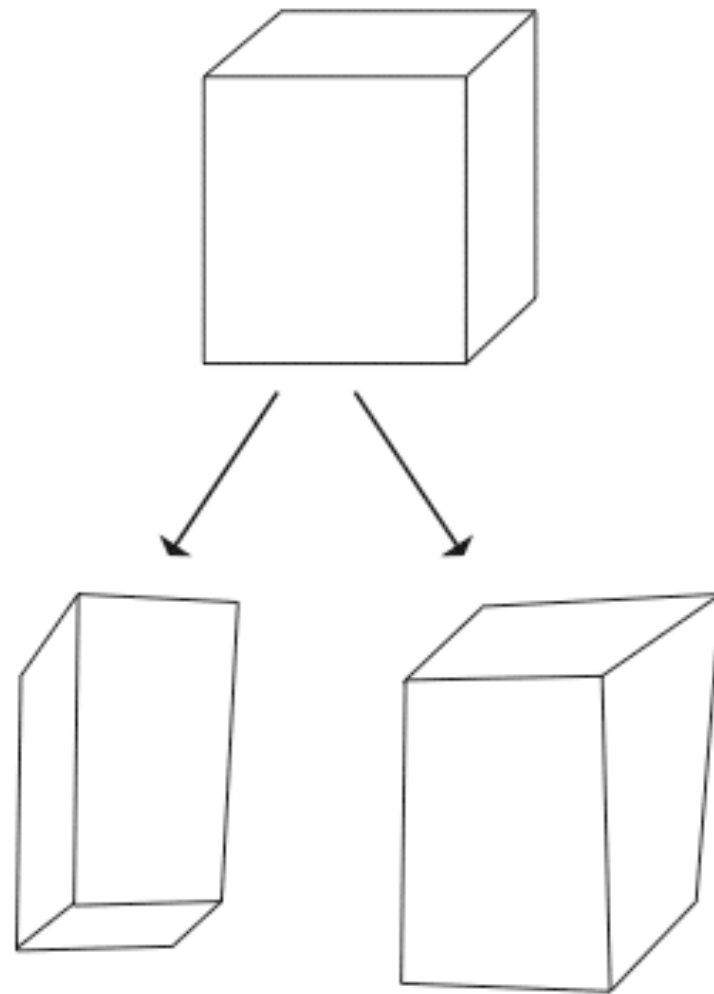
**Continually add vertices to make
corresponding boundaries have an
equal number**

Result of Triangulation

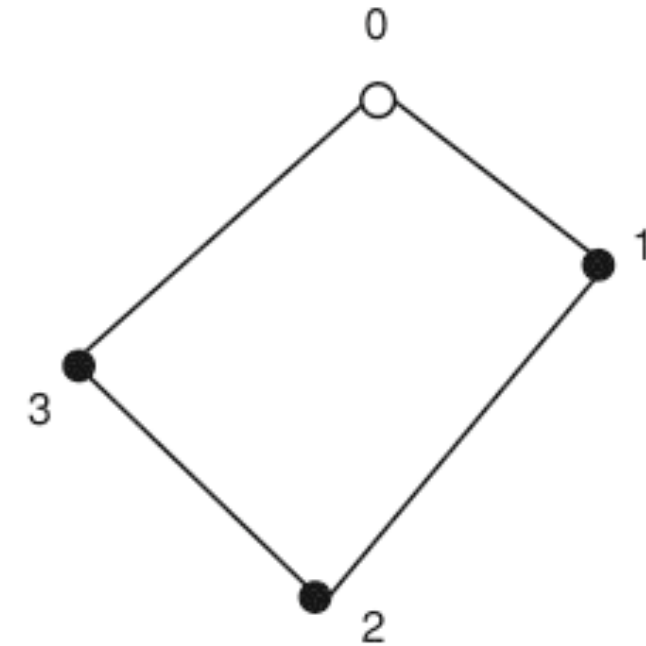
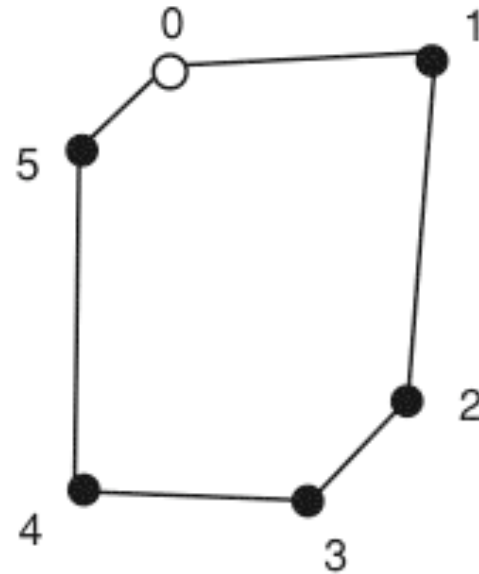
- Both meshes have the same triangulation on the mapped sphere

Other ways of creating a map

Object subdivision



Map the boundaries



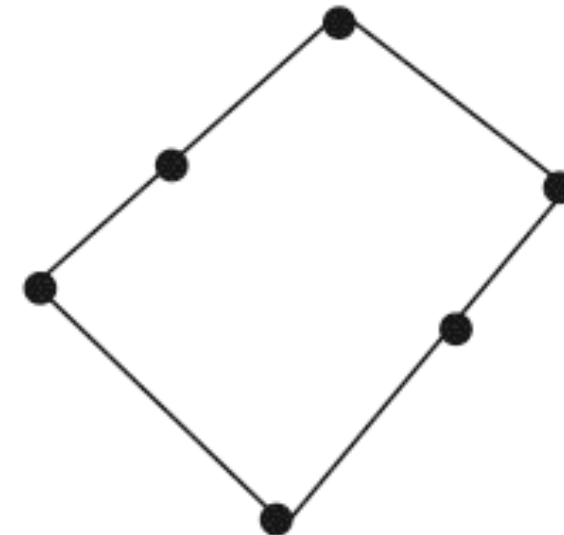
○ First vertex of boundary

Normalized distances

0	0.00
1	0.15
2	0.20
3	0.25
4	0.40
5	0.70

Normalized distances

0	0.00
1	0.30
2	0.55
3	0.70



Boundary after adding additional vertices

Summary

- Object interpolation
 - Creation of correspondence mapping
 - Interpolation

Image Morphing



Morphing

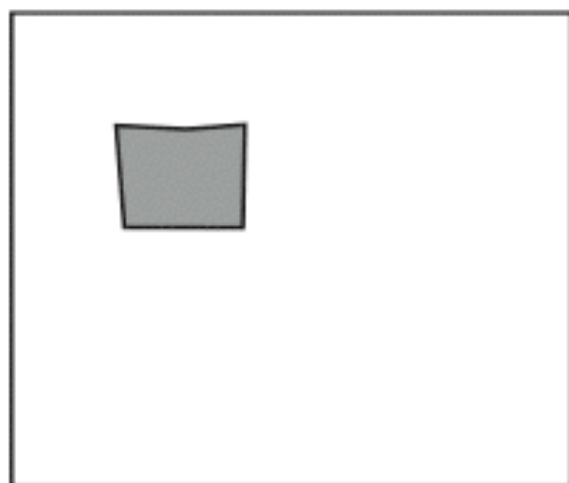


Image *A*

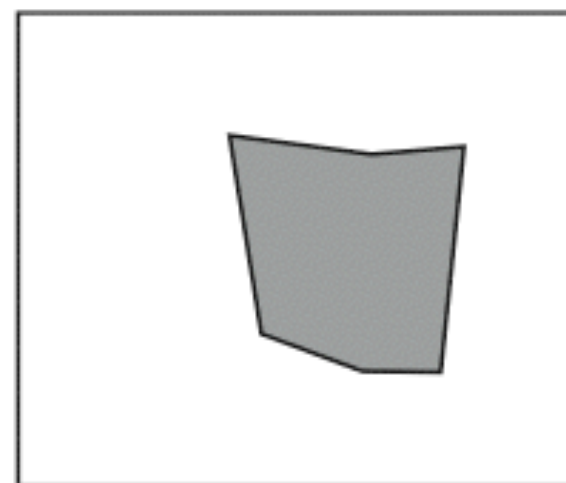


Image *B*

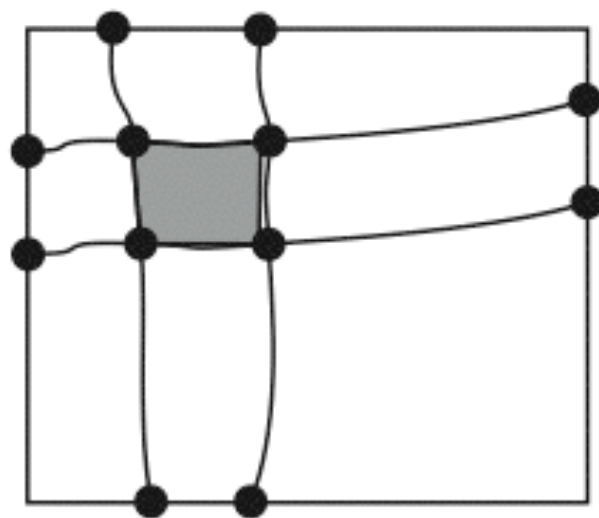


Image *A* with grid points and curves defined

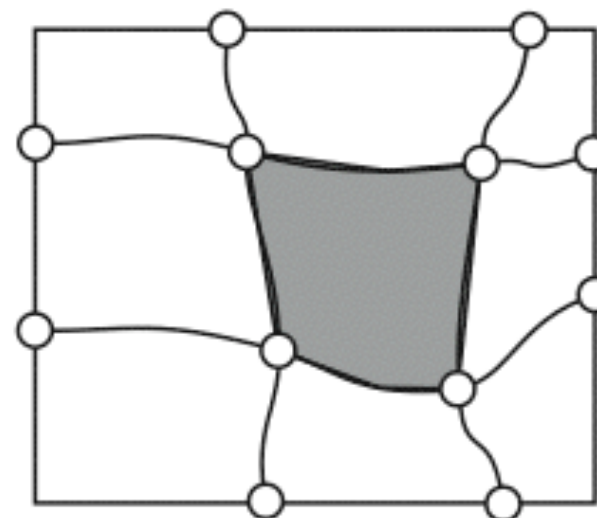
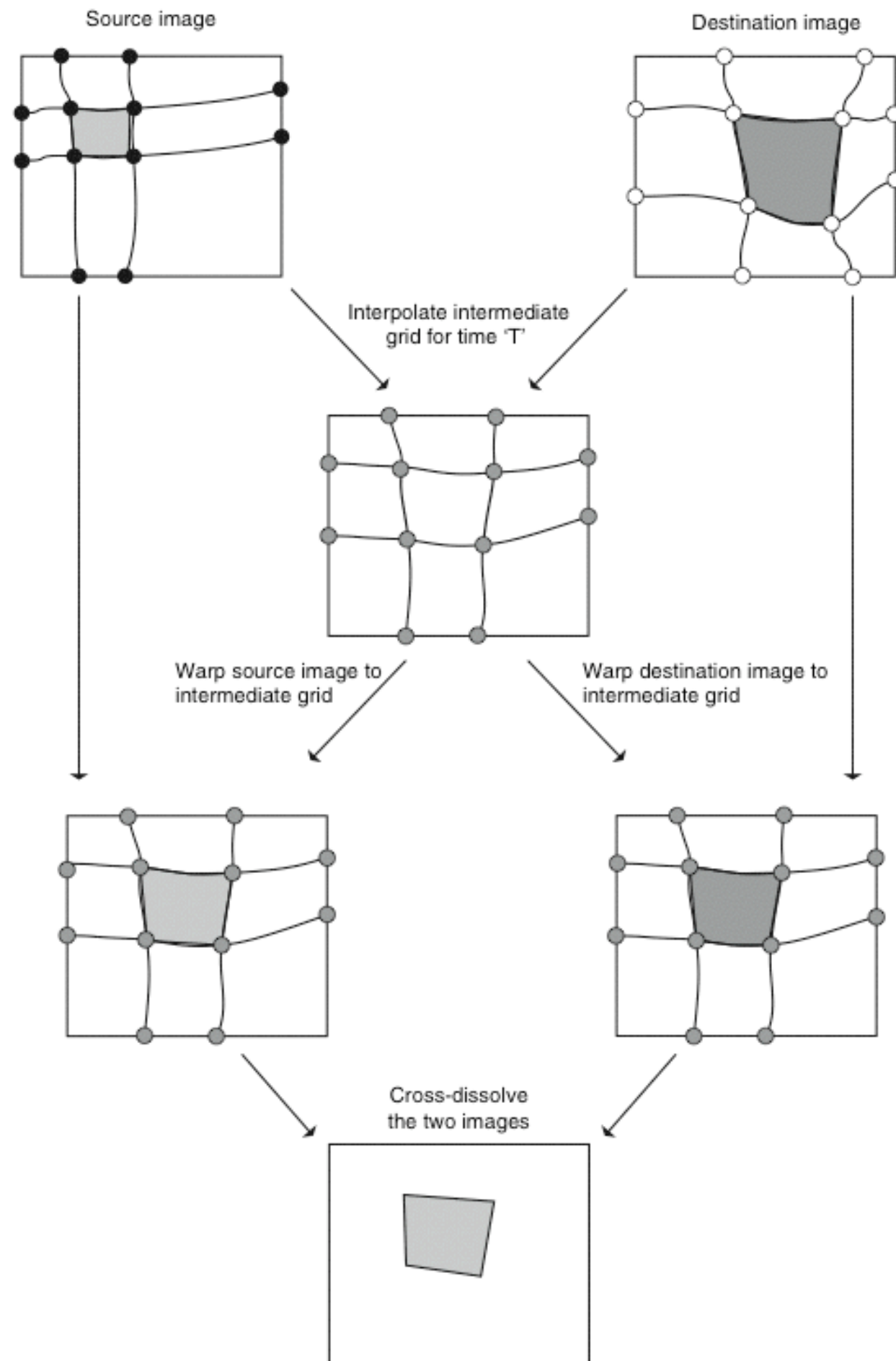
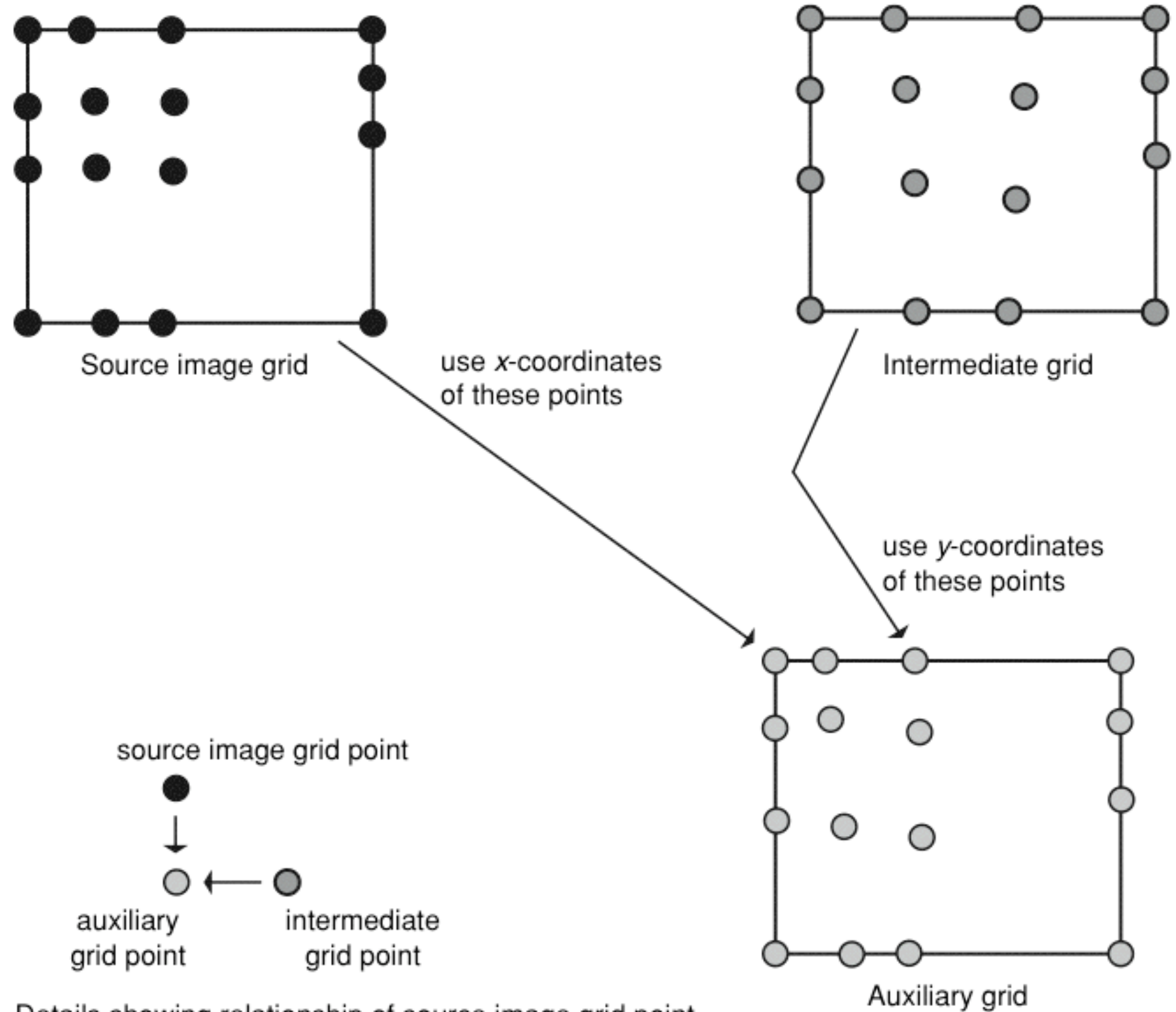


Image *B* with grid points and curves defined

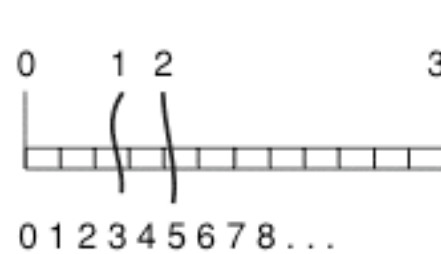
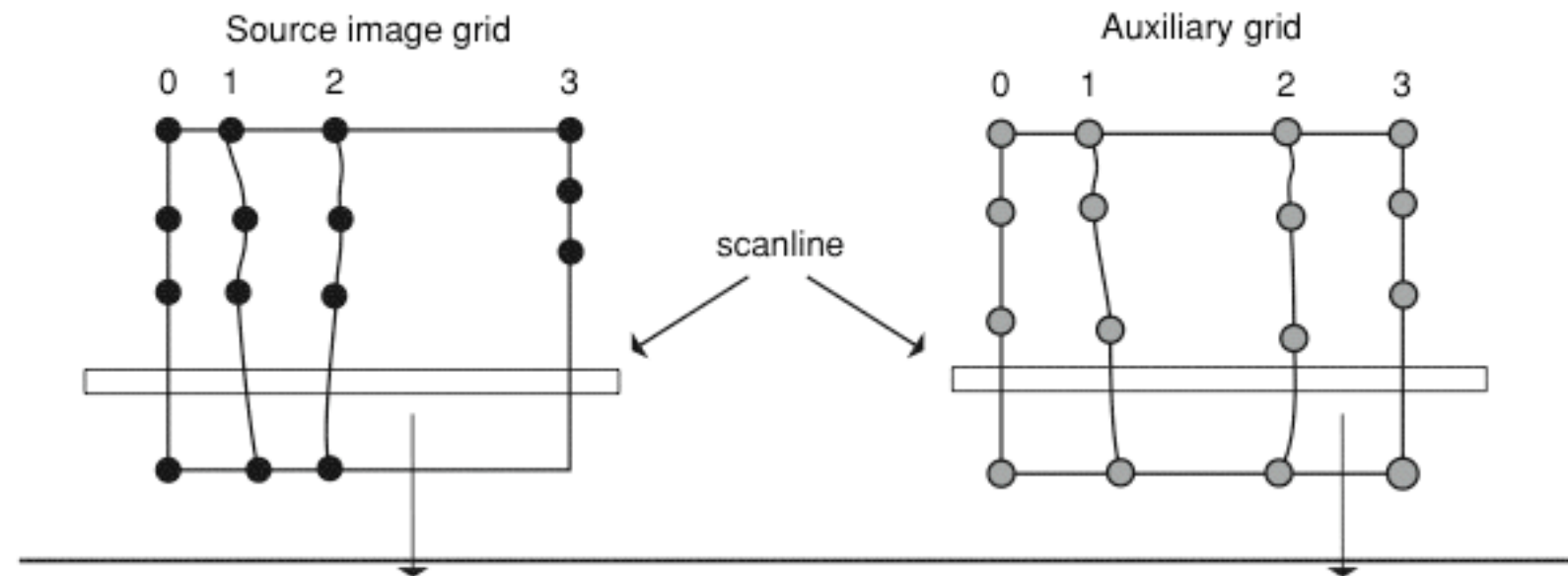
Morphing



Morph

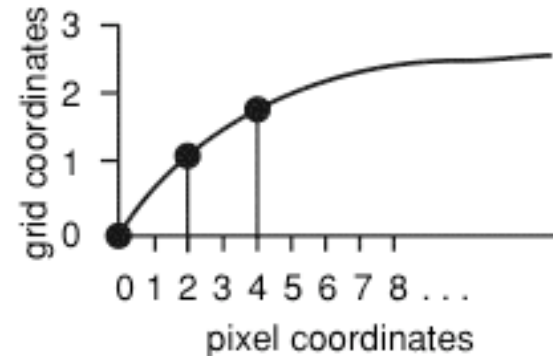
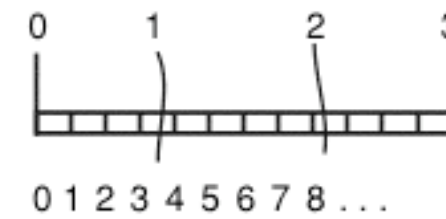


Morphing

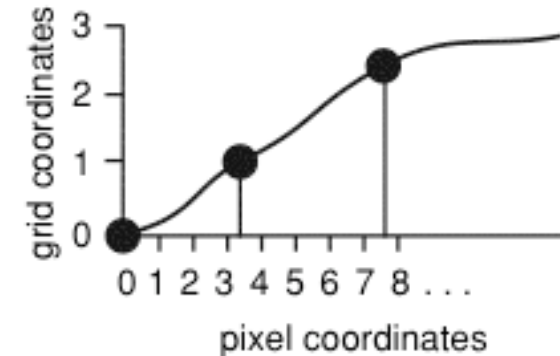


grid coordinates

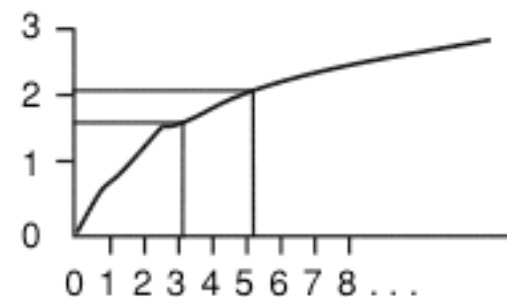
pixel coordinates



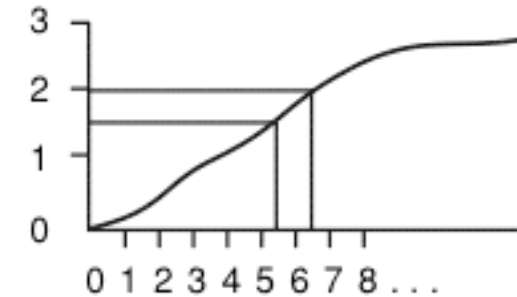
pixel coordinate to grid coordinate
graph for source image



pixel coordinate to grid coordinate
graph for auxiliary image

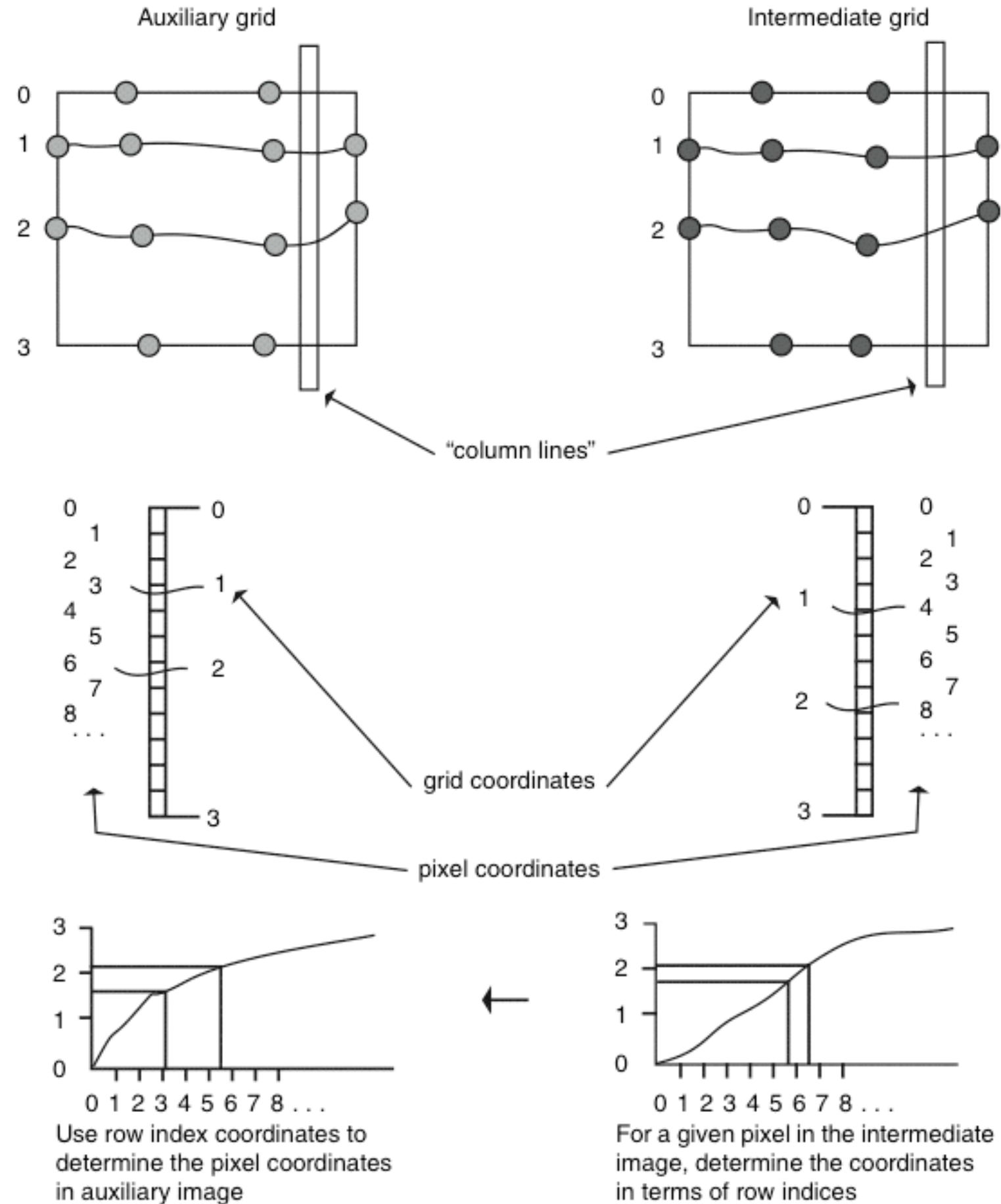


Use the graph to see where the column
indices map to image pixels. (Here, half of
pixel 3 and all of pixels 4 and 5 are useful)

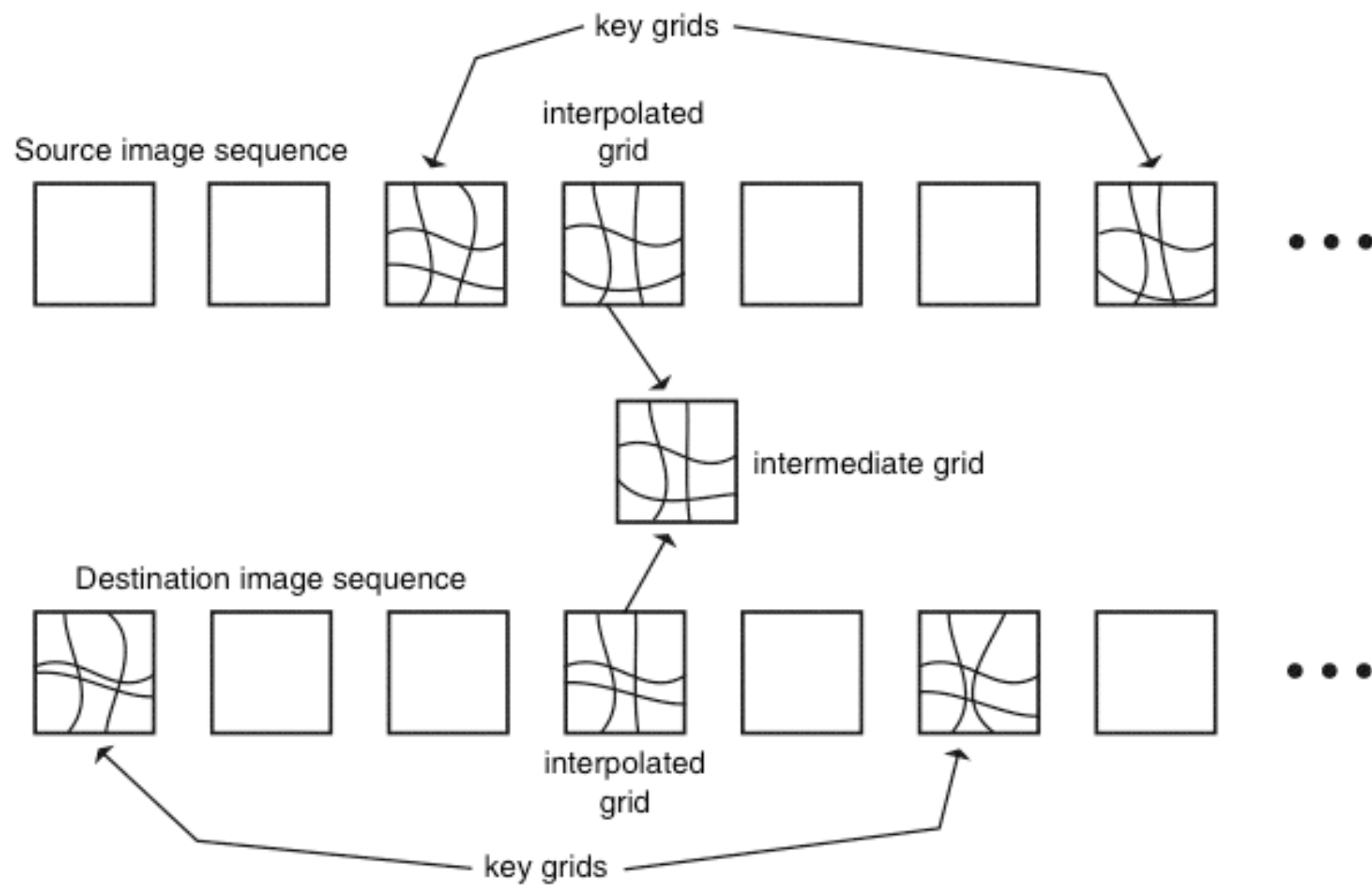


Use the graph to determine the image pixel's
range in terms of the column indices
(pixel 6 is shown)

Morphing

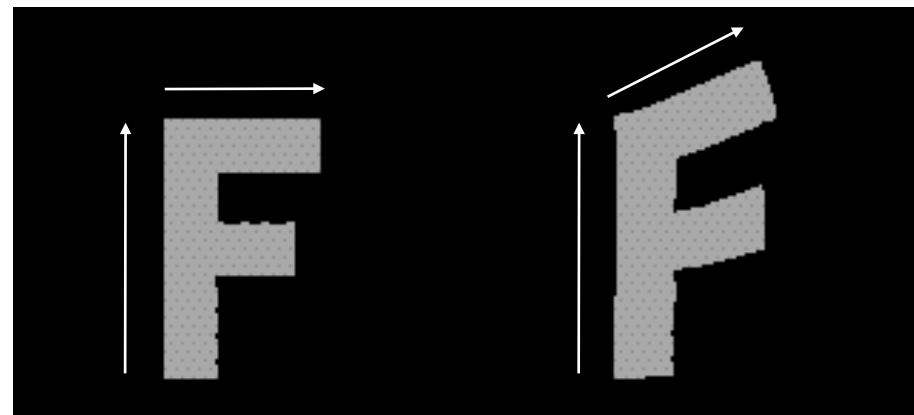


Morphing



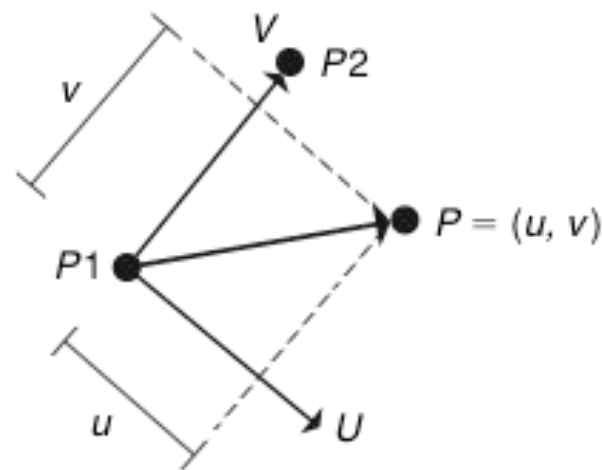
Morphing: feature based

**Given: corresponding user-defined
feature lines in source and
destination images**

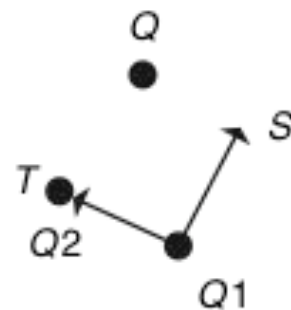


Morphing: feature based

Locate each pixel relative to each feature line in source and destination images



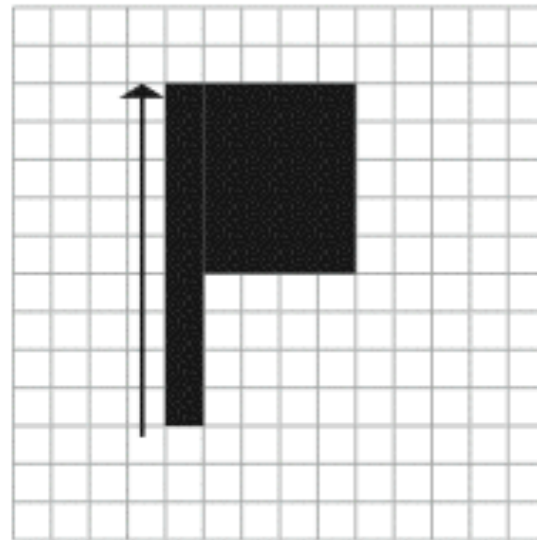
$$v = (P - P_1) \cdot \frac{(P_2 - P_1)}{|P_2 - P_1|^2}$$
$$u = \left| (P - P_1) \times \frac{(P_2 - P_1)}{|P_2 - P_1|^2} \right|$$



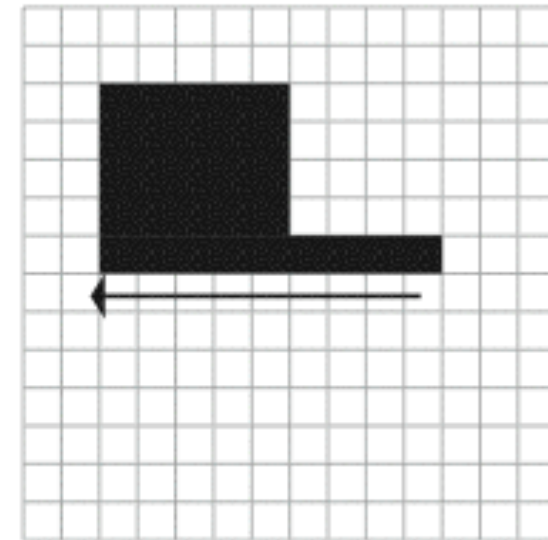
$$T = Q_2 - Q_1$$
$$S = (T_y, -T_x)$$
$$Q = Q_1 + uS + vT$$

Morphing: feature based

Source image and feature line

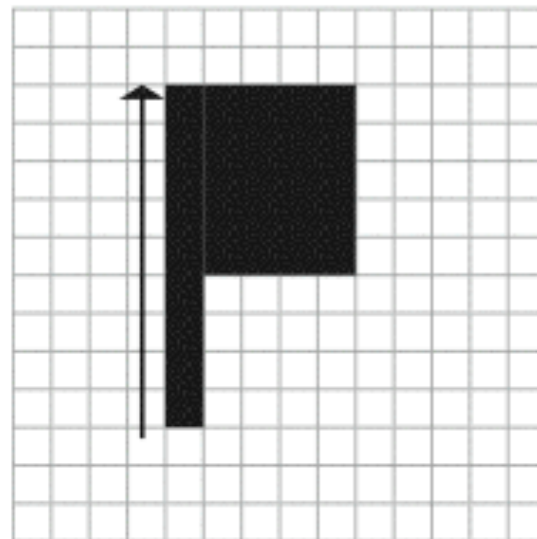


Intermediate feature line and
resulting image

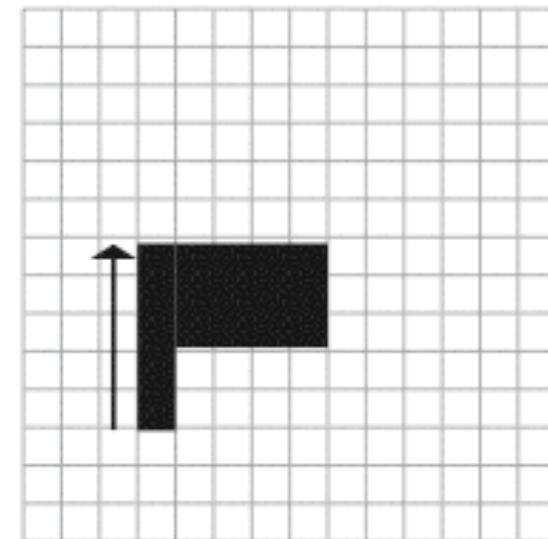


First example

Source image and feature line



Intermediate feature line and
resulting image



Second example

Summary

- Image morphing
 - Parameterizing images
 - Intermediate image
 - Auxiliary image
- Feature based image morphing

Summary

- Interpolation within an object
- Object interpolation (morphing)
- Image morphing